

Multimodal Emotion Recognition- a classifier based on EEG and eye movement recordings

Introduction

Emotions play an important role in our daily lives, specifically in human-human interactions: the emotional state of an individual modulates the way information is shared, affecting several external behaviours like posture, tone of voice, facial expressions and gestures, which in turn can alter the meaning of a sentence.

Moreover, the ability to recognize one's affective state has been shown to play an important role in learning, and is used to evaluate children development.

Introducing affective awareness in human computer interactions is currently the object of interest of many fields that go beyond psychiatry, with applications that encompass robotics, neuroscience, marketing, education and entertainment industries.

To overcome the limitations of purely behavioral cues (which can be deliberately masked), many studies turn to physiological signals. In particular, EEG offers a direct window into brain activity and is far less susceptible to voluntary control; features like Differential Entropy across the Delta–Gamma bands have proven to be strong predictors of affective states. Similarly, eye-tracking metrics (saccades, fixations, pupil dilation) capture attentional and arousal signals that complement EEG, providing a richer multimodal representation of emotion.

This project aims to develop a classifier for recognizing emotional state of an individual, discriminating between neutral state, happiness, sadness and fear, keeping as a reference the Emotion Meter framework proposed by Zheng et al.

The SEED IV Dataset

The SEED-IV dataset is an evolution of the original SJTU Emotion EEG Dataset (SEED), provided by the BCMI laboratory (Brain-like Computing and Machine Intelligence) in Shanghai.

It provides EEG signals collected with the 62-channel ESI NeuroScan System and eye movement features recorded by SMI eye-tracking glasses, which makes it a well-formed multimodal dataset for emotion recognition.

The experiment is organized in 3 sessions, performed in different days, each consisting of 24 trials per subject. This project will exploit only two sessions. During a trial, each of the 15 subjects (healthy, right-handed aged between 20 and 24 years) is shown a different clip which induces one of the 4 target emotions: happiness, sadness, fear, neutral state, while eeg activity and eye movements were recorded. After each clip, participants self-reported the emotion they experienced by selecting from the keywords.

EEG Dataset

The raw EEG signals of each subject for each trial are provided in a 62xN matrix, where the number of rows refers to the channels and N is a variable number of samples which depends on the duration of the trial considered.

Each recording is sliced into 4-second nonoverlapping segments. Each segment is regarded as one data sample. The final dataframes for each session amount to about 12000 samples each.

Each sample of raw eeg data underwent bandpass filtering between 1-45Hz and afterward, power spectral density (PSD) and differential entropy (DE) features were extracted at 5 frequency bands: delta: 1~4 Hz; theta: 4~8 Hz; alpha: 8~14 Hz; beta: 14~31 Hz; and gamma: 31~45 Hz.

During our initial data exploration, we organized the 62 EEG electrodes into six anatomical regions—Prefrontal, Frontal, Central, Parietal, Temporal, and Occipital—to manage redundancy and focus on key

brain areas. Dimensionality reduction via t-SNE failed to reveal distinct emotion-specific clusters, suggesting that the raw feature space remains highly overlapping. A correlation analysis further highlighted that many channel-band combinations are strongly interrelated, making it difficult to visually isolate the most informative signals. Guided by prior work pointing to the beta and gamma bands as particularly relevant, we computed ANOVA F-scores for each frequency band and channel region. The scores reinforced the importance of high-frequency activity and led us to select the three bands—beta, gamma, and delta—with the highest discriminative power for our classification task. The end result is a single table where each row corresponds to one trial and each column represents a specific feature. In the EEG portion, columns follow the naming convention <feature>_<region of interest>_<band> (for example 'power_Frontal_Theta').

EEG only classification

The 'eeg_only_top3.ipynb' notebook implements an EEG-only emotion-recognition pipeline using the top-3 ROI features

We load the top-3 ROI EEG features (features_eeg_top3.xlsx), extract the labels (y) and subject IDs (groups), and form our feature matrix (X). Using Leave-One-Subject-Out (LOSO) cross-validation on Session 1, we tune two classifiers (an SVM and a Random Forest) searching over C, kernel, and γ for the SVM, and tree-based hyperparameters for the RF. After selecting the best parameters, we retrain each model on all Session 1 data and save them (svm_model.joblib, rf_model.joblib). Finally, we cross-session evaluate by applying these retrained models to Session 2 data to measure how well our EEG-only pipelines generalize over time.

- Hyperparameter tuning (LOSO CV):
 - SVM: $C \in [10^{-3}, 10^2]$, kernel $\in \{\text{linear}, \text{rbf}\}$, $\gamma \in \{\text{scale}, \text{auto}, 0.01, 0.1\}$.
 - Random Forest: $n_estimators=50-200$, $max_depth=5-30$, $min_samples_split=2-10$, balanced class weights.
- Cross-session test: Retrained models are tested on Session 2 (same subjects, different recording day) to assess robustness against session drift
- Performance results show the SVM achieving a higher LOSO CV mean accuracy (33.6 %) compared to the Random Forest (30.9 %); the former was ultimately chosen as final model

The LOSO-CV results for the optimized SVM pipeline on Session 1 yielded an overall accuracy of 33.6% and a macro-averaged F1-score of 33.4 %. Class-level performance varied notably.

The SVM performs best on the neutral and happy states (classes 0 and 3) and struggles most with sadness (class 1), as evidenced by the lowest recall (22.2 %). These results highlight the need for strategies capable of boosting recall for more ambiguous classes.

Eye Dataset

SEED-IV provides also eye movement features recorded by SMI eye-tracking glasses. They are organized per subject and per trial. Each feature matrix has 31 rows (types of features) and N columns (time windows). Data are then reorganized in a data frame of 12765 rows (samples) and 34 columns: 31 features and three columns for subject, number of trial and target emotion. The 31 features are:

- Mean, standard deviation and differential entropy in four bands of the pupil diameter (X and Y)
- Mean and standard deviation of the dispersion (X and Y)
- Mean and standard deviation of the fixation duration (ms)
- Mean and standard deviation of saccade duration (ms) and saccade amplitude (°)
- Event statistics (blink frequency, fixation frequency, maximum fixation duration, total fixation dispersion, maximum fixation dispersion, saccade frequency, average saccade duration, average saccade amplitude, average saccade latency)

From an exploratory analysis of the dataset ('eye_movement_dataset_exploratory_analysis.ipynb' Jupyter Notebook), what emerges is that the class distribution is slightly imbalanced ('SAD' category is more represented, 'FEAR' is less represented), however there is an equal number of samples for each subject, ensuring subject-level balance. In a PCA 2D projection it's possible to see that there are overlapping characteristics across most classes (class 'HAPPY' shows some degree of separability), while t-SNE projection reveals some clusters related to nonlinear relationships in the data. The correlation matrix reveals the presence of strongly correlated features, indicating possible redundancy, but also uncorrelated features that might carry unique independent information.

Eye movement only classification

The notebook 'eye_movement_classification_nested_and_hybrid.ipynb' explores different supervised machine learning models to classify emotional states based on eye movement features.

First, to select the proper model, three classifiers are developed: SVM, Random Forest and kNearestNeighbors.

Each model is wrapped in a pipeline consisting of:

- Scaler: StandardScaler()
- Feature Selector:
SelectKBest(score_func=f_classif, k=grid_search.best_params_["select__k"]),
- Classifier:
 - SVC(kernel="rbf", class_weight="balanced")
 - RandomForestClassifier(n_estimators=50, random_state=42, max_depth=6, min_samples_split=2, class_weight="balanced")
 - KNeighborsClassifier(n_neighbors=5)

Starting from the Support Vector Machine, a GridSearchCV on full LeaveOneSubjectOut split is performed, with:

```
param_grid = {"select__k": [10, 15, 20, 25], "svc__C": [0.8, 0.9, 1.0, 1.2], "svc__gamma": [0.015, 0.017, 0.02]}
```

The final model is selected based on macro-averaged F1 score and the best parameters are identified. The performance of F1-macro (LOSO) is ~0.616, with k = 25, C = 0.8 and gamma = 0.015.

Keeping the same k, Random Forest and kNN are evaluated, but the performances are worse.

The models are tested on the second session to generalize over time: SVM is still the best one, it maintains generalization across sessions indicating strong cross-session robustness.

Chosen the SVM, also Nested Cross-Validation is performed to avoid information leakage in hyperparameter tuning. In an outer loop there is the subject-level LOSO split, while in the inner loop the GridSearchCV is applied to training folds only, excluding the test subject, to get a more reliable estimate of generalization performance. For each test subject there is an inner grid search run using LeaveOneGroupOut on remaining subjects, the best model is trained and evaluated on held-out subject and the best parameters and scores are saved per subject.

The final model was constructed by aggregating optimal hyperparameters across all subjects from the nested cross-validation phase:

- k is the mode (most frequent value) of selected features from all subject-level models
- C is the mean of optimal C values (SVM regularization parameter)
- gamma is the mean of optimal gamma values (RBF kernel width)

This aggregation strategy balances per-subject tuning outcomes and generalizes them into a single robust configuration. The model is trained on the entire first session dataset and evaluated on the second session dataset. The final pipeline is saved in "final_model_nested.joblib".

The best performance is observed for class 2 (Fear) with $F1 = 0.7361$, the lowest performance for class 3 (Happy), due to confusion with adjacent emotions. Class 1 (Sad) shows low precision and recall, indeed there is high confusion between class 1 and other classes (especially with class 0 and class 3). Classes 1 and 3 are underestimated, class 0 is overpredicted.

To improve the performances of the model, an hybrid strategy of feature selection is developed.

In the previous model, the features are selected based on the `f_score`: method used to rank and select relevant features based on their scores that represent the ratio of variability between groups (explained variance) to the variability within groups (unexplained variance). It evaluates features individually, so it's a univariate method. Now also `mutual_info_classif` is considered to capture non-linear relationships (MI can identify dependencies that correlation might miss), provide a clear ranking of feature importance based on their relevance to the target and identify and reduce redundancy between features.

So an hybrid feature selection strategy was applied by combining the normalized scores from both ANOVA F-test (`f_classif`) and mutual information (`mutual_info_classif`). Looping over a list of candidate values for `k` (optimal number of top-ranked hybrid features: [22, 25, 28, 29, 30, 31]), LOJO cross-validation is performed and the model with fixed `C` and `gamma` (previously obtained from nested cross-validation) is fitted and evaluated according to `f1_macro`. The best value of `k` is `k = 30`.

The final model is saved as "final_model_best_k_hybrid.joblib". It's fitted on `X_final` (first session dataset with only the selected features) and tested on the second session (completely independent).

On the test set, the final model achieved: accuracy: 57.88%, macro F1-score: 0.5573, F1 per class: 0.627 (class 0), 0.413 (class 1), 0.750 (class 2), 0.440 (class 3).

Compared to the baseline SVM model with `SelectKBest` ($F1 \approx 0.537$), the hybrid-selected model improved generalization by approximately +2.0% in macro F1, without sacrificing interpretability.

These results confirm that hybrid feature selection combined with tuned feature dimensionality leads to a more robust and generalizable model, particularly effective in multi-subject classification settings.

Modality Fusion

Modality fusion is a Machine learning technique that integrates different types of data (called modalities) in order to produce improved predictive performances with respect to those derived from the individual modalities. Depending on the level at which the integration happens, we can identify two main types of modality fusion:

- **Early Fusion:** raw data from different modalities is processed and merged into a single feature set, which is then used for training the model
 - allows for a comprehensive representation of the data
 - simple approach
 - single training process
 - suffers from high dimensionality of the data
- **Late Fusion:** individual models are trained on separate modalities, and their predictions are combined at a later stage according to a Fusion Rule
 - allows flexibility thanks to modularity
 - reduced dimensionality
 - base models can be optimized individually
 - loss of inter-modality information
 - higher complexity

Early Fusion Model

The two datasets (`eye_movement_dataset` and `features_eeg_roi`) are concatenated into a larger dataset: `df_fusion_roi`. To select the model, the following classifiers were tested:

- Support Vector Machine with RBF kernel:
SVC(kernel='rbf', C=10, gamma=0.001, class_weight='balanced')
- Random Forest:
RandomForestClassifier(n_estimators=200, max_depth=20, random_state=42)
- Logistic Regression: LogisticRegression(max_iter=1000, class_weight='balanced')

All models are evaluated using Leave-One-Subject-Out (LOSO) cross-validation to simulate generalization to unseen individuals and multiple feature selection strategies:

- No feature selection (baseline)
- SelectKBest (univariate ANOVA)
- L1-based selection (e.g., using Lasso)
- Random Forest-based feature importance

Each pipeline includes data normalization via StandardScaler.

The best model was SVM with SelectKBest, so after having identified with Optuna which could have been the best parameters, a Nested GridSearchCV LOSO is performed to tune the hyperparameters. The final model is saved in "best_svm_nested_grid_losa.joblib":

- k = 67
- C = 8
- gamma = 0.00075
- Nested LOSO Mean Accuracy: 62.07%

To conclude the analysis, the effect of varying the number of selected features on classification performance is investigated, using a SelectKBest feature selection strategy, while keeping the classifier and its hyperparameters C and gamma fixed. In this case, SMOTE is introduced in the Pipeline to avoid class unbalancing. The best number of features is 29. The final pipeline with k=29 and the same values of C and gamma is fitted on the whole dataset of the first session and evaluated on the second session. The model is saved in "early_fusion.joblib". The final accuracy is ~ 55%, class 2 consistently shows strongest classification performance, supported by both F1 and AUC. Class 1 is the most challenging, with lowest recall and F1. It's possible to conclude that feature selection matters, because using the right number of features (k=29) improves generalization without overfitting. Even moderate AUCs for Classes 0 and 3 indicate performance is better than chance and likely usable in a real-world system.

Late Fusion Model

To implement a learned late-fusion strategy, the project adopts a stacking architecture- a two level ensemble technique where the first level (base) models, one for each modality, are fed their respective data, and their outputs are used as inputs of a second level model, called "meta-learner".

The 'late_fusion' notebook implements a two-stage stacking ensemble that combines EEG and eye-tracking classifiers into a learned late-fusion meta-model. First, Session 1 EEG (features_eeg_top3.xlsx) and eye-tracking (eye_movement_reordered.xlsx) data are loaded and aligned by subject, trial, and label. Two base pipelines are instantiated with the same hyperparameters settled on in the unimodal models.

Using Leave-One-Subject-Out on Session 1, each pipeline is trained on all-but-one subject and their out-of-fold class-probabilities are collected. These probability vectors are horizontally concatenated per sample to form the meta-feature matrix (meta_X, meta_y).

Next, two meta-models (SVM and logistic regression) are tuned via GridSearch CV (LOSO CV, scoring = f1_macro) over hyperparameter grids, and their best estimators are selected. Both meta-models are then re-trained on the full Session 1 meta-dataset.

Between the two meta-learners, performance is highly comparable:

- LOSO CV (Session 1): Meta-SVM achieved a mean macro-F1 of 0.592 ± 0.128 , while Meta-LR scored 0.586 ± 0.128 —virtually identical generalization across held-out subjects.

- Hold-out Test (Session 2): Meta-LR holds a slight edge (56.1 % accuracy, macro-F1 = 0.550) over Meta-SVM (55.7 % accuracy, macro-F1 = 0.542), with the largest per-class gains on the “sad” and “happy” categories ($\Delta F1 \approx + 0.02$).

Given the Wilcoxon $p = 0.42$ (n.s.) on their LOSO F1 distributions, these small test-set differences are not statistically significant, so either meta-learner can be chosen with confidence.

Models Comparison

The notebook ‘models_comparison.ipynb’ evaluates how different data modalities and fusion strategies impact the performance of the models for emotion recognition, with a focus on both accuracy and F1-score across cross-validation and test sessions.

It’s possible to conclude that EEG-only underperforms and is least reliable: it has the lowest median accuracy in cross-validation and the narrowest interquartile range, so the performance is consistently poor.

Eye-only is inconsistent because mean accuracy is comparable to fusion methods, however, variance is high, indicating strong dependence on subject/session conditions. It shows potential but lacks robustness without fusion.

Both fusion methods stabilize performance: they achieve similar or slightly higher median accuracy than eye-only. The variance is significantly reduced, improving reliability.

Based on statistical test, no significant difference in F1-score between early and late fusion ($p = 0.73$) is appreciated and no significant difference between fusion models and eye-only models ($p > 0.5$).

Therefore, eye-tracking is the dominant modality, and EEG adds little to no improvement when fused with eye-based features. Fusion strategies do not significantly improve performance but help reduce variability.

Literature Comparison

In this project, LOSO CV on Session 1 is truly subject-independent. The final evaluation on Session 2 uses the same subjects as Session 1, but different trials: so, this is cross-session, not cross-subject. It’s more challenging than intra-session testing because it captures session-related variability (e.g., sensor drift, fatigue, noise).

EmotionMeter¹ uses a 16/8 train/test split within the same session. Thus, it’s a subject-dependent, intra-session setting (same subjects, seen in both train and test). Results are expected to be higher, but less generalizable to new subjects or recording sessions.

The EEG-only model in this project reaches 33.6%, which is lower than Zhong et al.’s 37.99% and potential reasons could be a less aggressive or different preprocessing pipeline, a more limited hyperparameter search space, particularly for C or an inaccurate preprocessing of EEG raw data.

Conclusion

In this project three classification pipelines are developed and evaluated for multimodal emotion recognition using EEG signals and eye movement data: EEG-only, eye-only, multimodal fusion (early and late). Despite achieving lower absolute performance compared to some studies in literature, our results provide valuable insights into the real-world feasibility and generalization of such models.

Limitations of this project are due to the fact that EEG and eye-tracking signals are inherently noisy and susceptible to session variability, fatigue, and sensor placement issues, which contribute to performance drops—especially in cross-session testing. Moreover, the parameter grid for models like

¹ Wei-Long Zheng, Wei Liu, Yifei Lu, Bao-Liang Lu, and Andrzej Cichocki, EmotionMeter: A Multimodal Framework for Recognizing Human Emotions. IEEE Transactions on Cybernetics, Volume: 49, Issue: 3, March 2019, Pages: 1110-1122, DOI: 10.1109/TCYB.2018.2797176.

SVM was constrained to reduce computational cost. Broader search spaces could potentially yield better results.

The use of nested cross-validation and Leave-One-Subject-Out (LOSO) strategies ensures no data leakage and unbiased estimates of model performance. Additionally, testing on an independent second session and unseen trials from the same subjects simulates deployment in real-world conditions where intra-subject variability and temporal drift are common. The use of interpretable feature selection methods (e.g., SelectKBest, Random Forest importance) allows to understand which features contribute most to classification. Finally, extensive comparisons across modalities (EEG-only, eye-only, early fusion, late fusion) and across evaluation levels (intra-session, cross-session) are performed, offering a clear understanding of the strengths and limitations of each approach.

This project could be improved with better EEG processing, broader searching space for hyperparameter tuning, richer feature engineering, advanced fusion methods or with the use of neural networks (very common in literature).